외계행성 대기
Exoplanetary Atmospheres

Lecture 6: Radiative Transfer IV — Line Formation, Scattering, and Monte Carlo
2026 October 19 (Monday)

Based on: Chapter 6 of Seager, *Exoplanet Atmospheres* (2010)

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Outline – Lecture 6

Line Formation and Radiative Transfer Solutions

An Introduction to Line Formation

Solutions I: Transmission and Thermal Emission

Solutions II: Scattering Atmospheres

Monte Carlo Radiative Transfer

Summary

Continuum, absorption lines, emission lines

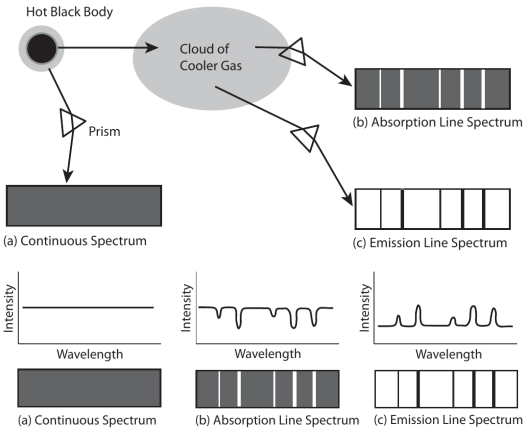


Figure 1: Origin of the three kinds of spectra (top) and what they look like over a narrow wavelength range (bottom). Seager (2010), Figs. 6.2 and 6.3.

- **Continuous spectrum:** a black body; the spectrum depends only on its temperature.
- **Absorption lines:** a cooler gas in front of the hot black body takes "bites" out of the continuum at the wavelengths its atoms and molecules absorb. The absorbed energy is re-emitted isotropically, so little of it returns to the line of sight.
- **Emission lines:** the same gas seen *without* the hot background. Only the re-emitted photons reach us, and they appear at exactly those wavelengths.
- What matters is not the gas alone but the gas *compared with what lies behind it*.

A layered atmosphere: the temperature gradient decides

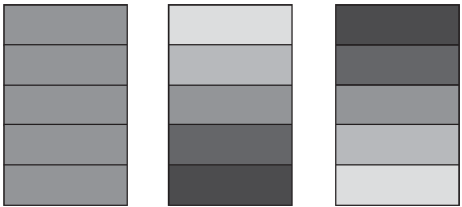


Figure 2: A five-layer atmosphere; darker means hotter. Left: isothermal. Middle: hot at depth, cooler above. Right: a temperature inversion. Seager (2010), Fig. 6.4.

Treat each layer as a homogeneous slab with its own temperature and the deepest layer as a black body.

- **Temperature decreasing outward:** every layer is a cool cloud in front of a hotter one $\Rightarrow$ an **absorption spectrum**. Each layer adds its own bites.
- **Temperature inversion** (hotter upward) $\Rightarrow$ an **emission spectrum**.
- **Isothermal:** no cooler gas in front, no hotter gas behind $\Rightarrow$ a black-body continuum with *no* features at all, however large the opacity.

The last point is counter-intuitive and worth proving (next section): spectral features measure temperature *contrasts*, not the mere presence of absorbers.

Transmission and scattered-light spectra

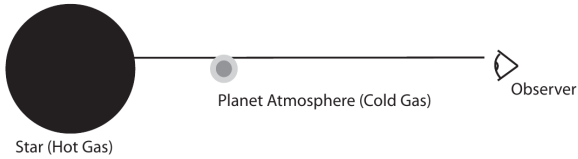


Figure 3: Origin of a transit transmission spectrum: hot stellar light through the cool planetary atmosphere. Seager (2010), Fig. 6.5.

- **Transmission:** the star is the hot black body, the planet's limb is the cool cloud ⇒ always an absorption spectrum. It is isolated by dividing the in-transit spectrum by the out-of-transit stellar spectrum.
- **Scattered light** (visible, near-IR): Seager prefers "scattered" to "reflected", because the light is refracted, diffracted and Rayleigh-scattered rather than bounced off a surface. To first order Earth's visible spectrum is sunlight that crossed the atmosphere *twice* (down to the ground or a cloud deck and back up), a weighted mix of clear and cloudy paths.

Isothermal Atmosphere: layer by layer

Take an n-layer LTE atmosphere whose deepest layer n is optically thick, so $I_n = B_n$. With Kirchhoff's law $\varepsilon_n = \kappa_n B_n$,

$$\mu \frac{\mathrm{d}I_n}{\mathrm{d}z} = \varepsilon_n - \kappa_n I_n = \kappa_n (B_n - B_n) = 0.$$

The intensity entering layer $n-1$ is therefore B_n, and in that layer

$$\mu \frac{\mathrm{d}I_{n-1}}{\mathrm{d}z} = \kappa_{n-1} (B_{n-1} - B_n) = 0 \qquad \text{because } T_{n-1} = T_n.$$

Repeating up to the top, $I_0 = \cdots = I_{n-1} = I_n = B$.

- Absorption and re-emission still happen in every layer, but they occur at the *same temperature* and hence at the same wavelengths: every photon removed is replaced by one of the same frequency.
- Note what the proof used: **only the equality of temperatures, not the size of κ_ν. A strongly absorbing and a weakly absorbing wavelength give the same $I = B$.**

Isothermal Atmosphere: from the formal solution

For an emitting atmosphere under **LTE conditions**, without scattering and with no radiation entering from above, the emergent intensity is

$$I(0,\mu,\nu) = \int_0^\infty B\big(T(\tau_\nu),\nu\big)\,\mathrm{e}^{-\tau_\nu/\mu}\,\frac{\mathrm{d}\tau_\nu}{\mu}.$$

(Recall: $\mu\,\mathrm{d}I/\mathrm{d}\tau_\nu = I - S_\nu$ gives $\mathrm{d}\big(I\mathrm{e}^{-\tau_\nu/\mu}\big)/\mathrm{d}\tau_\nu = -S_\nu \mathrm{e}^{-\tau_\nu/\mu}/\mu$; integrate from $\tau_\nu = 0$ to ∞, with $S_\nu = B_\nu$ in LTE without scattering.)

If T is constant, B leaves the integral, and $\int_0^\infty \mathrm{e}^{-\tau/\mu}\,\mathrm{d}\tau/\mu = 1$, so

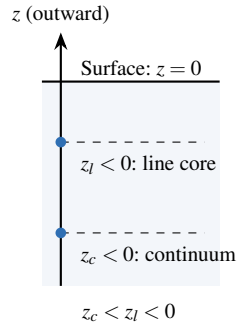
$$I(0,\mu,\nu) = B(T,\nu), \qquad F(0,\nu) = \pi B(T,\nu).$$

- The kernel $\mathrm{e}^{-\tau/\mu}/\mu$ is a *normalized weighting function*: the emergent intensity is a weighted average of B over depth, with weights concentrated near $\tau \sim \mu$.
- Different wavelengths have different opacities, so we see down to different depths at different wavelengths. But if the temperature is the same at every depth, every depth gives the same B, and all wavelengths look alike: no absorption or emission lines. Lines appear only when the temperature changes with depth.

Non-isothermal Atmosphere: a temperature gradient creates lines

Compare a wavelength in the line core (ν_l) with that in the nearby continuum (ν_c). By definition $\kappa(\nu_l) > \kappa(\nu_c)$.

- From $\mathrm{d}\tau_\nu = -\kappa_\nu \,\mathrm{d}z$ (z increases outward), the same value $\tau_\nu \approx 1$ is reached *higher up* in the line core than in the continuum: $z_c < z_l < 0$ ($|z_l| < |z_c|$).
- The emergent intensity is essentially B at $\tau_\nu \approx \mu$ (Eddington–Barbier approx.; slide 22). So the line core samples higher layers, the continuum deeper ones.
- **Temperature decreasing outward:** $T(z_l) < T(z_c)$, so $I(\nu_l) < I(\nu_c)$ — an **absorption line**.
- **Temperature increasing outward (inversion):** $T(z_l) > T(z_c)$, so $I(\nu_l) > I(\nu_c)$ — an **emission line**.



A spectrum across a line is therefore a *scan in altitude*: the wings, with lower opacity, look deeper, and the core looks highest.[1]

[1] Seager's text at this point reads "$I(\tau_\nu, \mu, \nu_l) < I(\tau_\nu, \mu, \nu_l)$"; the right-hand side should be the continuum, ν_c.

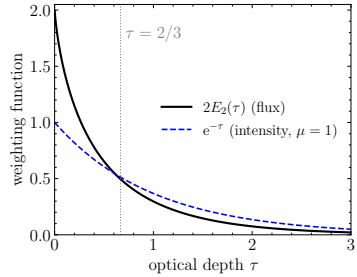
Non-isothermal Atmosphere: the emergent flux as a depth-weighted B_ν

Integrating the emergent intensity over the outgoing hemisphere,

$$F(0,\nu) = 2\pi \int_0^1 \int_0^\infty B(\tau_\nu,\nu)\, \mathrm{e}^{-\tau_\nu/\mu}\, \mathrm{d}\tau_\nu\, \mathrm{d}\mu = 2\pi \int_0^\infty B(\tau_\nu,\nu)\, E_2(\tau_\nu)\, \mathrm{d}\tau_\nu,$$

using $\int_0^1 \mathrm{e}^{-\tau/\mu}\, \mathrm{d}\mu = E_2(\tau)$, the exponential integral $E_n(x) = \int_1^\infty t^{-n} \mathrm{e}^{-xt}\, \mathrm{d}t$.

- $2E_2(\tau)$ is the **flux weighting function**: it is normalized ($\int_0^\infty 2E_2\, \mathrm{d}\tau = 1$), equals 2 at the top and falls roughly as $\mathrm{e}^{-\tau}/\tau$ at depth.
- This is the working formula for thermal emission from an exoplanet without scattering: a black body in each layer plus the (much harder) opacities. Most retrieval codes evaluate exactly this sum.
- Written per unit $\log p$, the integrand $E_2(\tau)\, \mathrm{d}\tau/\mathrm{d}\log p$ is the **contribution function**: it shows which pressures a given wavelength actually probes.



$2E_2(\tau)$ versus the intensity weighting $\mathrm{e}^{-\tau}$ ($\mu = 1$). Flux comes from slightly higher layers (mean depth $\tau = 2/3$ instead of 1): slanted rays reach $\tau_\nu/\mu = 1$ sooner.

Line formation in a toy atmosphere

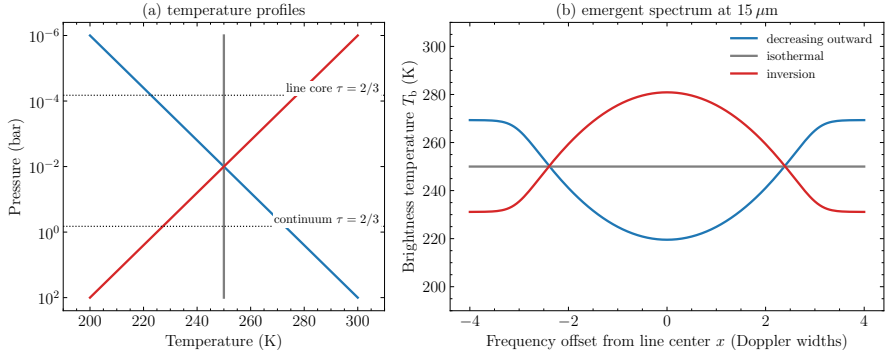


Figure 4: A line with a Gaussian opacity profile (peak 10^4 times the continuum) in three LTE atmospheres. Left: the temperature profiles and the pressures where $\tau = 2/3$ in the continuum and at line center. Right: the emergent flux as a brightness temperature. Computed with $F = 2\pi \int B E_2(\tau_\nu)\, \mathrm{d}\tau_\nu$; script `make_line_formation.py`.

The continuum reports T near 0.7 bar, the line core T near 10^{-4} bar; the isothermal case is featureless despite the huge opacity contrast.

Why a band has a floor: the minimum flux

A molecular absorption band is made of many closely spaced lines (Chapter 8 of Seager), so from here on we speak of an absorption *band*. Take an atmosphere whose temperature decreases outward.

- **The question:** what is the *minimum* flux in the band — and why is it not zero?
- There is always some radiation, however little, emerging at every wavelength.
- In LTE the size of the emergent flux depends on the temperature of the altitude where that flux *originates*.

So the question becomes: from which altitude does the flux at each wavelength come? The answer is set by the opacity, which varies strongly with wavelength — next slide. In an LTE atmosphere the flux at any wavelength is never zero; it is the Planck function of whatever layer that wavelength "sees".

Maximum and minimum flux: Earth's 15 μm CO_2 band

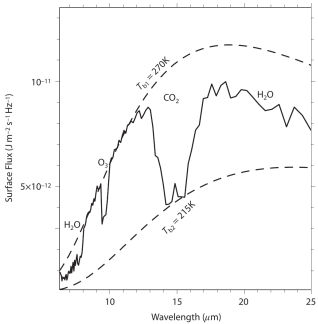


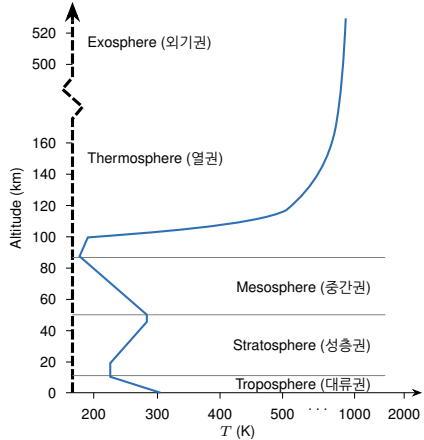
Figure 5: Earth's mid-infrared spectrum with black bodies at $T_{b1} = 270$ K and $T_{b2} = 215$ K. Reproduced from Seager (2010), Fig. 3.4.

- At a **low-opacity** wavelength the radiation can travel essentially unimpeded, so it emerges from *deep*, warm layers: this is the **maximum** flux, set by the surface or deep temperature.
- At a **high-opacity** wavelength the radiation is reprocessed in every layer on its way out; what finally escapes must have originated near the *top*, where it is cold: this is the **minimum** flux, set by the coldest layer the band can reach.
- Because opacity depends on wavelength, the spectrum *samples different temperatures* — in LTE, flux of different magnitude.
- **Earth:** between the bands the spectrum follows the warm $T_{b1} = 270$ K curve; the 15 μm CO_2 band bottoms out on the cold $T_{b2} = 215$ K curve, not at zero.
- The band depth therefore measures the *temperature range* between the continuum and band-center photospheres, not the amount of CO_2 as such. Brightness temperatures are a direct, if coarse, thermometer at different heights.

Reading Earth's temperature profile from its spectrum

- The deep O_3 (9.6 μm) and CO_2 (15 μm) absorption bands tell us that temperature decreases with height where they form: the **troposphere**, which holds most of the mass.
- At the very center of both bands there is a **narrow emission spike**. The core forms highest, so somewhere above the troposphere the temperature must rise again: the **stratospheric inversion**, heated by ozone absorbing solar UV.
- Why is the spike narrow? The opacity of a band scales with the number density of absorbers. Only at the band center is the opacity large enough for $\tau \approx 1$ to be reached in the low-density stratosphere; a little off-center, $\tau \approx 1$ drops back into the troposphere.
- **The narrowness of the emission core tells us that the emission features are forming in regions of low density.**

The width of an emission core thus encodes how high, and in how thin a layer, the inversion sits.



Earth's temperature profile. The exosphere extends to an altitude 500 km above Earth's surface. The ionosphere is a layer within the thermosphere. Redrawn after Seager (2010), Fig. 4.16.

Thermal inversions in hot Jupiters?

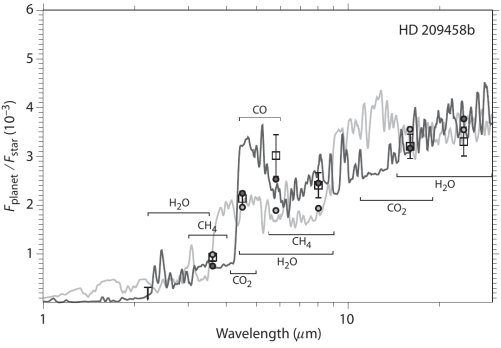


Figure 6: Historical interpretation of HD 209458b: Spitzer photometry (squares with error bars). Assuming H_2O and CO, the emission-only model (black) fits better than the absorption-only model (gray). Seager (2010), p. 112, Fig. 6.6; the inversion inference was revised by later observations.

- **Seager (2010):** secondary eclipses probe the permanent dayside of tidally locked hot Jupiters. Spitzer data suggested H_2O emission in some planets, absorption in others.
- **Earth:** weak emission cores within deep absorption bands indicate an inversion *above* the layers forming most of the absorption.
- **Initial HD 209458b interpretation:** strong emission without discernible absorption suggested an **on-average deep dayside inversion**, reaching the main feature-forming layers.
- **Update — HD 209458b:** HST/WFC3 resolved H_2O *absorption* (6.2σ). Together with Spitzer data, it supports **no inversion** over the probed $1–10^{-3}$ bar range (Line et al. 2016).
- **Update — WASP-121b:** resolved H_2O *emission* supports a stratospheric inversion (5σ; Evans et al. 2017).

Pure transmission: the Beer–Lambert law

In transmission there is neither emission nor scattering *into* the beam: $\varepsilon = 0$, hence $S = 0$, and

$$\mu \frac{\mathrm{d}I}{\mathrm{d}z} = -\kappa I \quad \Longrightarrow \quad I(z_f) = I(z_i) \exp\left[-\frac{1}{\mu} \int_{z_i}^{z_f} \kappa \, \mathrm{d}z' \right],$$

i.e. the intensity decays exponentially with the optical depth traversed:

$$\boxed{I(\tau_\nu) = I(0)\, \mathrm{e}^{-\tau_\nu/\mu}} \qquad \text{(the Beer–Lambert law, also called Beer's or Lambert's law)}$$

Not to be confused with Lambert's *cosine* law for a diffusely reflecting surface (Lectures 4–5); the two share only the name of J. H. Lambert.

In a transit the relevant quantity is the ratio of transmitted to incident stellar light along one ray, the stellar intensity cancelling:

$$\frac{F_{\mathrm{S},p,\mathrm{trans}}(\nu)}{F_{\mathrm{S},*,\mathrm{trans}}(\nu)} = \mathrm{e}^{-\tau_\nu},$$

where τ_ν is now the optical depth *along the chord* through the limb.

- This is one ray. The observed transit depth sums rays at all impact parameters through the annulus, each with its own chord optical depth (Lecture 3: $\tau_{\mathrm{chord}} \approx \sqrt{2\pi R_p / H}$ times the vertical one).
- Pressure broadening matters at depth, which is why the sum over altitudes cannot be skipped in a real calculation.

A transmission spectrum: HD 189733b

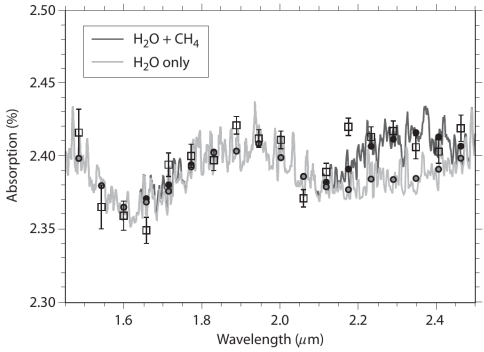


Figure 7: HST/NICMOS transmission spectrophotometry of HD 189733b (open squares with error bars). Curves: H_2O only (gray) or $H_2O + CH_4$ (black). Open circles and filled circles are, respectively, the H_2O-only and $H_2O + CH_4$ model predictions averaged over the observational wavelength bins, accounting for the instrument response. Seager (2010), Fig. 6.7.

- The signal is $\sim 5 \times 10^{-4}$ of the stellar flux, as estimated in Chapter 3: the area of an annulus about $5H$ tall relative to the stellar disk.
- Atoms and molecules detected in transmission in hot Jupiters include Na, K, H_2O, CO, CO_2 and (with JWST) SO_2.
- This particular CH_4 detection was later disputed: the NICMOS systematics had been underestimated (Gibson et al. 2011). A useful reminder that a feature is only as good as the instrument model behind it.

Seager's exercise 3(c) asks when thermal emission from the planet would matter in transmission: when the planet's own emission along the chord is not negligible compared with the transmitted starlight, i.e. for very hot planets observed in the infrared.

The emergent flux

The emergent surface flux follows from the flux definition (Chapter 2) applied to the outgoing hemisphere, $F = 2\pi \int_0^1 \mu\, I(0,\mu)\, \mathrm{d}\mu$ (no incoming radiation at the top in the thermal infrared):

$$\boxed{F(0,\nu) = 2\pi \int_0^1 \int_0^\infty S(\tau_\nu, \mu, \nu)\, \mathrm{e}^{-\tau_\nu/\mu}\, \mathrm{d}\tau_\nu\, \mathrm{d}\mu}$$

- If the source function $S(\tau,\mu,\nu)$ is *known*, the emergent intensity and the emergent flux — the planet's spectrum — can be computed directly from these two integrals. The μ integral of $\mathrm{e}^{-\tau/\mu}$ produces the exponential integral $E_2(\tau)$, so $F(0,\nu) = 2\pi \int_0^\infty S(\tau_\nu) E_2(\tau_\nu)\, \mathrm{d}\tau_\nu$ for an isotropic S.
- For $S = B(T(\tau))$ (pure absorption, LTE) this is the standard *forward model* of an emission spectrum: choose $T(\tau_\nu)$, i.e. a temperature–pressure profile and an opacity function, and integrate.
- In many cases, however, this straightforward solution is not possible. The main complication is that the source function itself depends on $I(\tau,\mu,\nu)$, the quantity we are trying to solve for.

Thermal emission: the formal solution with $S = B$

At infrared wavelengths molecular absorption dominates and scattering can be neglected: $\kappa = \alpha_{\rm abs}$ (absorption only), and in LTE $S_{\rm therm} = \varepsilon/\kappa = B(\tau_\nu, \nu)$. The equation $\mu\,\mathrm{d}I/\mathrm{d}\tau_\nu = I - B$ has the solution

$$I(0,\mu,\nu) = \int_0^\infty B(\tau_\nu,\nu)\,\mathrm{e}^{-\tau_\nu/\mu}\,\frac{\mathrm{d}\tau_\nu}{\mu}, \qquad \boxed{F(0,\nu) = 2\pi\int_0^1\int_0^\infty B\,\mathrm{e}^{-\tau_\nu/\mu}\,\mathrm{d}\tau_\nu\,\mathrm{d}\mu}.$$

- "Powerfully simple": each layer is a black body; the only hard ingredient is the opacity, which fixes $\tau_\nu(z)$.
- Exact in LTE without scattering, given $T(z)$. This makes it the workhorse for secondary-eclipse and directly imaged spectra.
- Thermal emission peaks near the Planck maximum: ~ 0.8–$1\,\mu$m for 2000–3000 K ultra-hot Jupiters, $\sim 15\,\mu$m for Earth, longer for colder planets.
- Next slide shows data for one example, the Spitzer and Hubble Space Telescope observations of HD 189733b.

An emission spectrum: HD 189733b

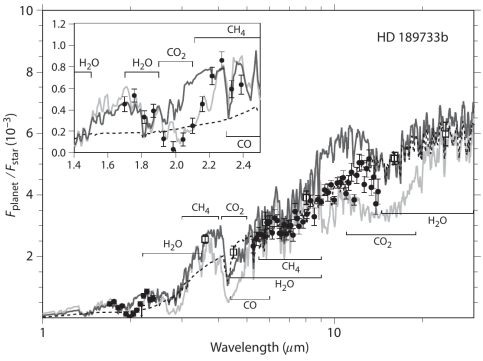


Figure 8: Secondary-eclipse data for HD 189733b: Spitzer photometry (squares), Spitzer/IRS (circles), HST/NICMOS (inset), with best-fit models. Seager (2010), Fig. 6.8.

- The planet/star contrast reaches $\sim 5 \times 10^{-3}$ in the mid-infrared, ten times the transmission signal.
- Order of magnitude (Chapter 3, Rayleigh–Jeans limit):

$$\frac{F_p}{F_*} \approx \frac{T_{\mathrm{b},p}}{T_{\mathrm{b},*}} \left(\frac{R_p}{R_*}\right)^2 .$$

  With $R_p/R_* = 0.155$ and $T_{\mathrm{b},*} \approx 5000\,\mathrm{K}$, a contrast of 3.4×10^{-3} at $8\,\mu\mathrm{m}$ implies $T_{\mathrm{b},p} \approx 700\,\mathrm{K}$; the full Planck ratio gives $\approx 1300\,\mathrm{K}$, close to the measured dayside value.
- Absorption features of H_2O and CO/CO_2: temperature decreasing with height on this dayside.

A constant source function

We can solve the RT equation under the assumption of a constant source function with altitude. Take S independent of depth between an upper level ($\tau_{\min}$) and a lower one ($\tau_{\max}$); along a vertical ray,

$$I(\tau_{\min}) = I(\tau_{\max})\,\mathrm{e}^{-(\tau_{\max}-\tau_{\min})} + S\left[1 - \mathrm{e}^{-(\tau_{\max}-\tau_{\min})}\right].$$

The first term is what survives of the light entering from below, the second what the slab emits along the path.

- **Optically thick slab** (deep in the atmosphere), $\tau_{\max} - \tau_{\min} \gg 1$: $I = S$. The incoming light is forgotten; only local conditions matter, and in LTE $I = S = B$: a featureless continuum. *Lines do not form at large optical depth.*

$$I(0,\nu) = S(\nu).$$

- **Optically thin top layer**, $\tau_{\min} = 0$, $\tau_{\max} < 1$, so $\mathrm{e}^{-\tau} \approx 1 - \tau$:

$$I(0,\nu) = I(\tau_{\max},\nu) - \tau_{\max}(\nu)\,[I(\tau_{\max},\nu) - S(\nu)].$$

  [Absorption Line Spectrum] The layer *removes* light if the background is brighter than its source function (i.e., $I(\tau_{\max},\nu) > S(\nu)$).

  [Emission Line Spectrum] The layer *adds* light if it is fainter (i.e., $I(\tau_{\max},\nu) < S(\nu)$), in proportion to its optical depth $\tau_{\max}(\nu)$.

Absorption or emission: the four cases

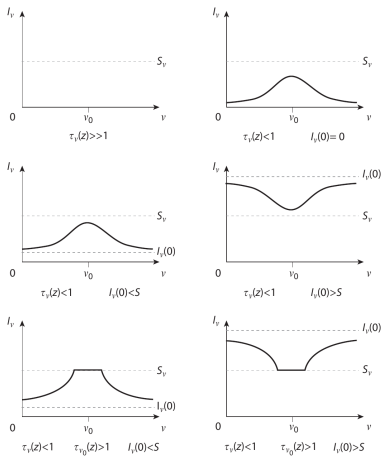


Figure 9: Line formation for a source function constant with depth and frequency. Seager (2010), Fig. 6.9, after Rutten.

Read the panels with $I(\tau_{\max}) \equiv I^0_\nu$, the light from behind, and $\tau_\nu(z)$ the layer's optical depth, largest at line center ν_0:

- **Top left** — thick at all ν: $I = S$, no line.
- **Top right** — thin, no backlight ($I^0_\nu = 0$): emission line, strength $\propto \tau_\nu$.
- **Middle left** — thin, $I^0_\nu < S$: emission line on top of the background.
- **Middle right** — thin, $I^0_\nu > S$: absorption line.
- **Bottom** — thick at line center only ($\tau_{\nu_0} > 1$): the core *saturates* at $I = S$, giving a flat-topped emission or flat-bottomed absorption line.

Absorption lines require light from behind that is *brighter* than the layer's own source function — the same statement as "cool gas in front of a hot body".

A linear source function: Eddington–Barbier approximation

Let $S(\tau_\nu) = a_\nu + b_\nu\,\tau_\nu$ (independent of μ). The formal solution gives exactly

$$I_\nu(0,\mu) = a_\nu + b_\nu\mu, \qquad F_\nu(0) = 2\pi\int_0^1 (a_\nu + b_\nu\mu)\,\mu\,\mathrm{d}\mu = \pi\left(a_\nu + \tfrac{2}{3}b_\nu\right). \qquad \boxed{F_\nu(0) = \pi S(\tau_\nu = 2/3).}$$

- The emergent **flux** is characteristic of $\tau_\nu \approx 2/3$; the **intensity** along the vertical ray, $I(0,\mu=1) = S(\tau_\nu = 1)$. This is the Eddington–Barbier relation.[2]
- It does not require the absence of scattering, only a source function that is (close to) linear in τ.
- Because τ_ν depends on frequency, so does the physical height of $\tau_\nu = 2/3$: this is the mathematical form of **"the line core forms higher (outer layer)"**.
- The same $I = a_\nu + b_\nu\mu$ gives $\boxed{K(\tau_\nu,\nu) = J(\tau_\nu,\nu)/3}$ for the second and zeroth moments — the **Eddington approximation**, which reappears as the third Eddington coefficient of the two-stream method.

  Inside the atmosphere, with $I = a_\nu + b_\nu\mu$ for all $-1 \le \mu \le 1$: $J = \frac{1}{2}\int_{-1}^{1} I\,\mathrm{d}\mu = a_\nu$ and $K = \frac{1}{2}\int_{-1}^{1} I\mu^2\,\mathrm{d}\mu = a_\nu/3$ (the odd $b_\nu\mu$ term drops out of both), so $K = J/3$. It fails at the surface, where $I(\mu<0) = 0$.

[2] Seager writes $F(0,\nu) = S(\tau_\nu = 2/3)$ and $F(0,\nu) = S(\tau_\nu = 1)$ (Eqs. 6.32–6.33); the first lacks the factor π, and the second is the vertical intensity rather than a flux.

A Scattering Atmosphere: Why scattering makes everything harder

Absorption and thermal emission tie the radiation to the *local* temperature. Scattering does not:

- A scattered photon can travel far before it is absorbed. The long-distance photon travel by scattering therefore **couples distant parts of the atmosphere**, and even the open boundary at the top influences the field deep inside.
- The source function now contains the radiation field itself,

$$S_{\text{scat}}(\tau,\mu,\phi) = \omega_0 \int_{4\pi} \mathscr{P}(\mu,\phi;\mu',\phi')\, I(\tau,\mu',\phi')\, \mathrm{d}\Omega',$$

  so the RT equation becomes an **integro-differential equation**: I appears on both sides, once differentiated and once integrated over all directions.
- Few analytic solutions exist; classical texts spend many pages even on isotropic scattering (Chandrasekhar 1960; Sobolev 1975).

Here ω_0 ($\tilde{\omega}$ in Seager) is the single-scattering albedo and $\mathscr{P} \equiv P/4\pi$ the single-scattering phase function, normalized as $\int_{4\pi} \mathscr{P}\, \mathrm{d}\Omega = 1$ (Lecture 3).

Direct beam and diffuse field

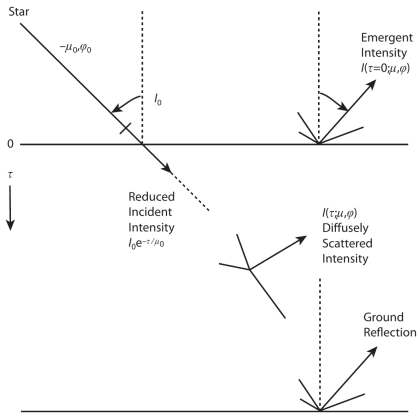


Figure 10: The attenuated direct beam from the star and the diffusely scattered intensity. Seager (2010), Fig. 6.10, after Chamberlain & Hunten.

Starlight arrives as a parallel beam from direction (μ_0, ϕ_0):

$$I_*(0,\mu,\phi) = I_0\,\delta(\mu-\mu_0)\,\delta(\phi-\phi_0).$$

The incident flux follows from angular integration (Seager, p. 120):

$$\begin{aligned} F_*(0,\mu,\nu) &= \int_0^{2\pi}\int_{-1}^{0} \mu I_0\,\delta(\mu-\mu_0)\,\delta(\phi-\phi_0)\,\mathrm{d}\mu\,\mathrm{d}\phi \\ &= \mu_0 I_0. \end{aligned}$$

In this signed-flux equation, incoming $\mu_0 < 0$; the incident-flux magnitude is $|\mu_0| I_0$. For attenuation below, μ_0 denotes the positive magnitude.

Unscattered, the beam is attenuated as

$$I_{\mathrm{dir}}(\tau) = I_0\,\mathrm{e}^{-\tau/\mu_0}\,\delta(\mu-\mu_0)\,\delta(\phi-\phi_0).$$

It is convenient to split the field into this **direct beam**, known analytically, and the **diffuse field** I of photons scattered at least once, which is what we must solve for.

Scattering RT: an angularly coupled problem

Let τ_ν increase inward and $\mu > 0$ denote outgoing rays. For coherent scattering, the source receives photons from all directions:

$$\boxed{\mu \frac{\mathrm{d}I_\nu(\tau_\nu,\Omega)}{\mathrm{d}\tau_\nu} = I_\nu(\tau_\nu,\Omega) - S_\nu(\tau_\nu,\Omega)}$$

$$S_{\mathrm{scat},\nu}(\tau_\nu,\Omega) = \omega_0 \int_{4\pi} \mathscr{P}_\nu(\Omega;\Omega') I_\nu(\tau_\nu,\Omega')\,\mathrm{d}\Omega'.$$

- $\Omega = (\mu,\phi)$, $\mathrm{d}\Omega' = \mathrm{d}\mu'\,\mathrm{d}\phi'$; $\mathscr{P}$ is normalized by $\int \mathscr{P}\,\mathrm{d}\Omega = 1$. Pure scattering has $\omega_0 = 1$.
- Scattering removes photons from one ray and adds photons from other rays. Unlike an LTE thermal source $B_\nu(T)$, the scattering source depends on the *unknown radiation field*.
- The equation is therefore **integro-differential**: every direction is coupled to the others.

Notation correction: use $I - S$, as in Seager Eq. 6.1; the source is an integral over solid angle, without a flux factor μ'. The printed Eqs. 6.38–6.40 have inconsistent signs/angular measures.

Separate the direct beam from diffuse scattering

Write $I_{\rm tot} = I_{\rm dir} + I_{\rm diff}$. Here $\mu_0 > 0$ is the *magnitude* of the incoming cosine, so the stellar beam travels in direction $\Omega_0 = (-\mu_0, \phi_0)$.

$$S_{{\rm single},\nu}(\tau_\nu, \Omega) = \omega_0 I_0 \, \mathrm{e}^{-\tau_\nu/\mu_0} \, \mathscr{P}_\nu(\Omega; \Omega_0),$$
$$S_{{\rm diff},\nu}(\tau_\nu, \Omega) = \omega_0 \int_{4\pi} \mathscr{P}_\nu(\Omega; \Omega') I_{{\rm diff},\nu}(\tau_\nu, \Omega') \, \mathrm{d}\Omega',$$

- The direct beam is known analytically. Its source term describes the **first scattering of stellar photons**; the diffuse term accounts for photons that scatter again.
- For an outgoing ray, a layer of total depth τ_b gives

$$\boxed{I_\nu(0, \mu) = I_\nu(\tau_b, \mu) \mathrm{e}^{-\tau_b/\mu} + \frac{1}{\mu} \int_0^{\tau_b} S_\nu(t, \mu) \mathrm{e}^{-t/\mu} \, \mathrm{d}t}$$

- This formal solution is still valid, but $S_{\rm diff}$ contains the unknown $I_{\rm diff}$: it must be solved self-consistently.

Seager Eqs. 6.41–6.42 rewritten with the required $1/\mu$ and an explicit lower-boundary term. For stellar scattering alone, $S = S_{\rm single} + S_{\rm diff}$; thermal emission adds $(1 - \omega_0) B_\nu$.

Single scattering in a thin layer

Optically thin cirrus or aerosol layers high in the atmosphere can be dominated by one scattering of the incident stellar beam. Set $S = S_{\text{single}}$ and neglect illumination from below:

$$I_\nu(0,\mu) = \frac{\omega_0 I_0 \mathscr{P}_\nu}{\mu} \int_0^{\tau_b} \mathrm{e}^{-t/\mu_0} \mathrm{e}^{-t/\mu}\, \mathrm{d}t = \frac{\omega_0 \mu_0 I_0 \mathscr{P}_\nu}{(\mu+\mu_0)} \left[1 - \mathrm{e}^{-\tau_b(1/\mu+1/\mu_0)}\right].$$

The exponentials account for attenuation **before** and **after** scattering. For $\tau_b/\mu \ll 1$ and $\tau_b/\mu_0 \ll 1$,

$$\boxed{R(\mu,\phi;\mu_0,\phi_0) \equiv \frac{\pi I_\nu(0,\mu,\phi)}{\mu_0 I_0} \simeq \frac{\pi\omega_0\tau_b}{\mu\mu_0} \mathscr{P}_\nu(\mu,\phi;-\mu_0,\phi_0)}$$

- R is Seager's **dimensionless bidirectional reflectance** (BRDF): a Lambert-normalized reflected intensity. On Earth, the BRDF can be determined from satellite measurements.
- At fixed geometry, albedo and phase function, $R \propto \tau_b$. No isotropic-scattering or two-stream assumption is needed.

Seager Eq. 6.45 correction: the incoming attenuation is e^{-t/μ_0}, not $\mathrm{e}^{-t}\mu_0$. The finite-depth expression is exact for the single-scattering term, not for unrestricted multiple scattering.

Single scattering: when the thin-layer approximation holds

The exact single-scattering result is

$$R = \frac{\pi\omega_0 \mathscr{P}_\nu}{\mu + \mu_0}\left[1 - \mathrm{e}^{-\tau_b(1/\mu + 1/\mu_0)}\right].$$

- For the approximation $R \simeq \pi\omega_0\tau_b\mathscr{P}_\nu/(\mu\mu_0)$ to hold, $\tau_b \ll 1$ alone is not enough. We need

  $$\frac{\tau_b}{\mu_0} \ll 1 \ \text{(incoming path)} \qquad \text{and} \qquad \frac{\tau_b}{\mu} \ll 1 \ \text{(outgoing path)},$$

  i.e. the *path* optical depths, not only the vertical depth τ_b, must be small.
- Near grazing angles (μ or $\mu_0 \to 0$) even a thin layer fails this, and R saturates at

  $$R \to \frac{\pi\omega_0 \mathscr{P}_\nu}{\mu + \mu_0},$$

  independent of τ_b (the Lommel–Seeliger law), instead of growing as $1/(\mu\mu_0)$.
- Once the path depth exceeds unity, multiple scattering adds to this value, the more so the closer ω_0 is to 1.

Multiple scattering and Earth's spectrum

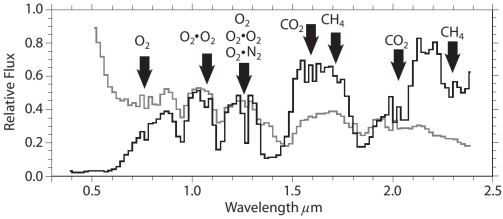


Figure 11: Earth as an exoplanet. Both spectra were observed by telescopes on Earth, using the Moon as a reflector. Gray (earthshine): Sun → Earth (scattering/reflection) → Moon → telescope on Earth. **Black (lunar eclipse): Sun → Earth's atmospheric limb (transmission/refraction) → Moon → telescope on Earth.** The long slant paths in transmission and the cloud/surface-weighted paths in scattered light produce different band strengths. Labels identify O_2, O_2–O_2 and O_2–N_2 collision-induced absorption, CO_2 and CH_4. Relative fluxes emphasize spectral shapes, rather than absolute planet/star contrasts. Seager (2010), Fig. 6.11; Pallé et al. (2009).

- Once repeated scattering matters, the source and radiation field are coupled; a simple single-scattering solution is no longer sufficient.
- **Adding–doubling:** calculate reflection and transmission for a thin homogeneous layer. Combine layers, including repeated reflections between them; doubling builds a thick slab efficiently.
- Different homogeneous layers can be added to represent a **vertically inhomogeneous** atmosphere.
- Two hot Jupiters had visible geometric-albedo upper limits below 0.15, compared with Jupiter's ~ 0.5; this helped explain the difficulty of scattered-light detection.
- Earth's scattered and transmitted spectra probe different paths and weight clouds and gases differently. Their molecular-band strengths therefore differ (left figure).

Radiative transfer in cloudy atmospheres

The same RT equations apply to clouds. The numerical difficulty is the **sharp opacity change at an optically thick cloud boundary**. Seager (p. 125) summarizes several practical treatments:

1. **Cloud top as a boundary:** place the effective atmospheric surface at the top of an opaque cloud and prescribe its emerging radiation, instead of solving through the cloud interior.
2. **Partial cloud coverage:** combine cloudy and clear columns, possibly with different cloud-top heights:

$$\boxed{I_{\text{disk}} \simeq f_{\text{cl}} I_{\text{cloudy}} + (1 - f_{\text{cl}}) I_{\text{clear}}}.$$

   Here f_{cl} is the projected-area cloud fraction; the columns are assumed radiatively independent.
3. **Smooth the boundary:** replace an abrupt cloud edge with an exponential density decrease for numerical convenience.
4. **Approximate cloud scattering analytically:** prescribe the radiation emerging from a cloud layer without solving its full internal transfer.

For realistic, nonuniform **3D cloud structures**, **Monte Carlo RT** approach is the most accurate way to deal with scattering from clouds.

Monte Carlo RT: follow photon packets

Traditional RT solves for the intensity field; Monte Carlo RT samples **individual photon-packet histories**. Their ensemble estimates the same radiation field.

- A packet represents many photons at frequency ν, with a position, direction and energy; polarization can also be tracked.
- Draw the next interaction distance, absorption or scattering, and the new direction from the appropriate probability laws. Follow each packet until it is **absorbed or escapes**.
- **Strength:** irregular geometry and spatially varying properties, including clumpy 3D clouds, can be treated directly.
- **Cost:** many histories are needed to reduce sampling noise; optically thick regions require many scattering events. Seager discusses simulations with millions of packets or more.

Scope here: time-independent, monochromatic, coherent scattering. The method can be applied to 3D media, thermal emission and temperature-structure calculations.

Sampling a free path: inverse cumulative probability

Let $p(x)$ be a normalized probability density on $[a,b]$. Draw ζ uniformly in $(0,1)$ and invert its cumulative distribution:

$$\boxed{\zeta = \int_a^{x_0} p(x)\,\mathrm{d}x, \qquad \int_a^b p(x)\,\mathrm{d}x = 1}.$$

The probability of no interaction after a path optical depth t is e^{-t}. Thus the next-interaction probability density is

$$p(t)\,\mathrm{d}t = \mathrm{e}^{-t}\,\mathrm{d}t, \qquad t \geq 0.$$

$$\zeta = 1 - \mathrm{e}^{-t} \quad \Longrightarrow \quad \boxed{t = -\ln(1-\zeta)}.$$

- t is optical depth *along the packet's ray*, not vertical depth. Convert it into a distance s using the local extinction: $t = \int_0^s \chi_v(\mathbf{r} + s'\hat{\mathbf{n}})\,\mathrm{d}s'$.
- In a homogeneous medium, $s = t/\chi_v$. If a boundary is closer than the sampled interaction, the packet reaches that boundary first.

The photon-packet algorithm

1. **Initialize.** Launch a packet from the stellar source; set its entry position, direction, frequency and energy.
2. **Propagate.** Sample $t = -\ln(1-\zeta_1)$ and find the corresponding path length. If the packet exits first, record it; otherwise move to the interaction point.
3. **Choose the interaction.** With a new random number, scatter if $\zeta_2 < \omega_0$ and absorb otherwise:

$$\boxed{\Pr(\text{scatter}) = \omega_0, \qquad \Pr(\text{absorb}) = 1-\omega_0}.$$

4. **Scatter and repeat.** Draw a new outgoing direction from $p(\Omega \mid \Omega') = P(\Omega;\Omega')/(4\pi)$; update polarization if included, then return to step 2. Coherent scattering preserves ν.
5. **Absorb or escape.** Absorption deposits the packet energy and ends its contribution to this monochromatic scattering calculation. Escaping packets contribute to the emergent radiation.

Absorption does not destroy energy: re-emission at thermal wavelengths is outside this scattering-only example. A surface encounter requires an appropriate absorption/reflection boundary condition.

From escaping packets to observables and errors

Bin escaped packets by direction, exit location and frequency. Their energy, normalized to the injected luminosity and bin geometry, gives the emergent flux; directional bins describe phase and inclination dependence (and polarization, if tracked).

- **Equal-energy packets:** if each packet carries energy $E_{\rm pkt}$ over a normalization time Δt, then $L_{\rm bin} = N_{\rm bin} E_{\rm pkt}/\Delta t$. Conversion to intensity or observer flux must include the relevant area, solid angle and distance.
- **Sampling error:** Seager uses Poisson counting errors. For independent, equal-weight packets and well-populated bins,

$$\boxed{\sigma_N \simeq \sqrt{N_{\rm bin}}, \qquad \frac{\sigma_F}{F} \simeq \frac{1}{\sqrt{N_{\rm bin}}}}.$$

  Halving the statistical error requires roughly four times as many packets in that bin.
- **Numerical check:** in a steady scattering-only run, escaped energy plus absorbed energy should match injected energy.
- Monte Carlo naturally handles complex clouds, but noise and long random walks make optically thick calculations expensive. Weighted-packet methods require their own variance estimates.

Take-home messages

Line formation. Spectral features measure temperature contrasts: an optically thick isothermal LTE atmosphere emits a featureless black body. Opaque wavelengths form high, transparent ones deep; temperature falling outward gives absorption, an inversion gives emission. Emergent flux $\approx \pi S(\tau_\nu = 2/3)$.

Solutions. Transmission obeys Beer's law along the chord; thermal emission is $F = 2\pi \int B E_2 \, \mathrm{d}\tau$ without scattering.

Scattering and Monte Carlo. Scattering couples directions; thin layers give $R \propto \omega_0 \tau P/(4\mu\mu_0)$. Clouds require boundary treatments or explicit scattering calculations. Sample photon paths and interactions; escaped packets give observables with counting noise $\propto N^{-1/2}$.

Next. Approximate solutions (Rosseland, Eddington, the grey atmosphere) and the vertical thermal structure of an atmosphere (Lecture 7).

References

- Seager, S. 2010, *Exoplanet Atmospheres: Physical Processes* (Princeton University Press), Chapter 6.
- Heng, K. 2017, *Exoplanetary Atmospheres: Theoretical Concepts and Foundations* (Princeton University Press), Chapter 3.
- Chandrasekhar, S. 1960, *Radiative Transfer* (Dover).
- Mihalas, D. 1978, *Stellar Atmospheres*, 2nd ed. (Freeman).
- Sobolev, V. V. 1975, *Light Scattering in Planetary Atmospheres* (Pergamon).
- Madhusudhan, N. and Seager, S. 2009, *ApJ* **707**, 24; Line, M. R., et al. 2016, *AJ* **152**, 203; Evans, T. M., et al. 2017, *Nature* **548**, 58 — thermal inversions.
- Swain, M. R., et al. 2008, *Nature* **452**, 329; Gibson, N. P., et al. 2011, *MNRAS* **411**, 2199.
- Pallé, E., et al. 2009, *Nature* **459**, 814.
- Code, A. D. and Whitney, B. A. 1995, *ApJ* **441**, 400 — Monte Carlo scattering.